

# Commands used to generate logs

Matteo Franzil, Valentino Armani, Luis Augusto Dias Knob

2024-05-28

## Contents

|          |                                                                                                             |           |
|----------|-------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Ignored logs</b>                                                                                         | <b>1</b>  |
| 1.1      | Ignored namespaces and users . . . . .                                                                      | 1         |
| 1.2      | Endpoints different than api/apis . . . . .                                                                 | 1         |
| 1.3      | API Server watching objects . . . . .                                                                       | 1         |
| 1.4      | kube-controller-manager watching objects . . . . .                                                          | 2         |
| 1.5      | kube-controller-manager getting and creating tokens for GC and RQ controllers . .                           | 2         |
| 1.6      | Kube Scheduler watching objects . . . . .                                                                   | 3         |
| 1.7      | Nodes watching objects . . . . .                                                                            | 3         |
| 1.8      | Nodes creating tokens . . . . .                                                                             | 4         |
| 1.9      | Nodes getting their own status . . . . .                                                                    | 4         |
| 1.10     | Nodes patching their status to update conditions . . . . .                                                  | 4         |
| 1.11     | CoreDNS watching namespaces . . . . .                                                                       | 4         |
| 1.12     | Kube-proxy watching nodes . . . . .                                                                         | 4         |
| <b>2</b> | <b>Log setups level 1</b>                                                                                   | <b>4</b>  |
| 2.1      | Namespace log N1 - Simple namespace creation and deletion . . . . .                                         | 4         |
| 2.2      | Namespace log N2 - Namespace with resource quotas . . . . .                                                 | 5         |
| 2.3      | Pod log P1 - Simple Pod creation and inspection . . . . .                                                   | 6         |
| 2.4      | Pod log P2 - Image download . . . . .                                                                       | 7         |
| 2.5      | ConfigMap C1 - Create a ConfigMap and use it in a Pod, then delete it . . . . .                             | 8         |
| 2.6      | Secret S1 - Create a Secret and use it as environment in a Pod, then delete it . . . . .                    | 9         |
| 2.7      | Secret S2 - Create a Secret and mount it in a Pod . . . . .                                                 | 10        |
| 2.8      | Node log N01 - Node inspection . . . . .                                                                    | 11        |
| 2.9      | Certificate log CSR1 - Create a CertificateSigningRequest and approve it . . . . .                          | 11        |
| 2.10     | Role log R1 - Create a Role and bind it to a User . . . . .                                                 | 12        |
| 2.11     | Role log R2 - Create a ClusterRole and bind it to a Group . . . . .                                         | 13        |
| 2.12     | Role log R3 - Create a ServiceAccount for a long-running Pod and bind it to a Role . .                      | 14        |
| 2.13     | Access Review AR1 - Use can-i to view permissions of self, other users, over different namespaces . . . . . | 16        |
| 2.14     | Access Review AR2 - Use auth whoami to view the current user . . . . .                                      | 19        |
| 2.15     | Access Review AR3 - Impersonating a user . . . . .                                                          | 19        |
| <b>3</b> | <b>Log setups level 2</b>                                                                                   | <b>21</b> |
| 3.1      | Deployment log D1 - Create a Deployment for a ReplicaSet and expose it as a Service                         | 21        |
| 3.2      | Deployment log D2 - Scale a Deployment and update its image . . . . .                                       | 22        |
| 3.3      | Deployment log D3 - Roll back, pause and resume a Deployment . . . . .                                      | 23        |
| 3.4      | Deployment log D4 - Watch a Deployment fail and recover . . . . .                                           | 24        |
| 3.5      | ReplicaSet log RS1 - Use bare ReplicaSets to manage Pods . . . . .                                          | 25        |
| 3.6      | DaemonSet log DS1 - Manage a DaemonSet . . . . .                                                            | 26        |

|     |                                                       |           |
|-----|-------------------------------------------------------|-----------|
| 3.7 | Job log J1 - Manage a Job . . . . .                   | 28        |
| 3.8 | CronJob log CJ1 - Manage a CronJob . . . . .          | 28        |
| 4   | <b>Log setups level 3</b>                             | <b>29</b> |
| 4.1 | Stateful Set log SS1 - Manage a StatefulSet . . . . . | 29        |

## 1 Ignored logs

### 1.1 Ignored namespaces and users

The following namespaces are ignored:

- `falco`
- `kube-flannel`

The following users/SAs are ignored:

- `system:serviceaccount:kube-flannel:flannel`

### 1.2 Endpoints different than `api/apis`

We will ignore endpoints which are not the regular API ones.

- Verb: any
- Users: any
- Objects: any
- Endpoints: any other than `/api` or `/apis`

### 1.3 API Server watching objects

The API server monitors core components with a 10minute timeout. It remains unclear what the API server watches in other groups: we probably need to deploy more components to see what the API server watches.

- Verb: `watch`
- Users: `system:apiserver`
- Objects: `configmaps`, `clusterroles`, `namespaces`, `serviceaccounts`, `resourcequotas`, `clusterrolebindings`, `rolebindings`, `secrets`, `nodes`, `Pods`, `roles`
- Namespaces: none, apart from `configmaps` in `kube-system`. Additionally, if legacy service accounts are used, the API server also watches `kube-apiserver-legacy-service-account-token-tracking`, a CM in `kube-system`.

### 1.4 kube-controller-manager watching objects

The `kube-system` namespace watches objects, similar to the API server. However, `ConfigMaps` are watched over non-namespaced objects and also `certificateigningrequests` are watched.

I believe that the objects watched by the Kube Controller Manager are specific to what is deployed in the cluster. For example, if the cluster has a deployment, the Kube Controller Manager will watch also deployments.

- Verb: `watch`

- Users: `system:kube-controller-manager`
- Objects: same as the API server, plus `certificateigningrequests`
- Namespaces: none

## 1.5 kube-controller-manager getting and creating tokens for GC and RQ controllers

The kube-controller-manager gets every less than an hour the serviceaccounts `/api/v1/namespaces/kube-system/serviceaccounts/generic-garbage-collector` and `/api/v1/namespaces/kube-system/serviceaccounts/resourcequota-controller`. It then creates tokens for them, which expire in an hour.

### 1.5.1 1

- Verb: `get`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/generic-garbage-collector`

### 1.5.2 2

- Verb: `get`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/resourcequota-controller`

### 1.5.3 3

- Verb: `create`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/generic-garbage-collector/token`

### 1.5.4 4

- Verb: `create`
- Users: `system:kube-controller-manager`, groups `system:authenticated`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/resourcequota-controller/token`

## 1.6 Kube Scheduler watching objects

The Kube Scheduler watches pods, nodes, namespaces, always with a 10-minute timeout.

- Verb: `watch`
- Users: `system:kube-scheduler`
- Objects: `pods`, `nodes`, `namespaces`
- Namespaces: none

### 1.6.1 Extension-apiserver-authentication

This object is also watched by the Kube Scheduler.

- Verb: `watch`
- Users: `system:kube-scheduler`
- Object: `configmaps/extension-apiserver-authentication`
- Namespaces: `kube-system`

## 1.7 Nodes watching objects

Each node in the cluster watches pods (non-namespaced), nodes (themselves), and some configmaps.

### 1.7.1 Pods

- Verb: `watch`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `pods`
- Namespaces: `none`

### 1.7.2 Nodes

- Verb: `watch`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `nodes/{node-name}`
- Namespaces: `none`

### 1.7.3 ConfigMaps

Every node hosting some Pod of any namespace will be in charge of watching the serviceaccounts of the Pods that are running.

The CMs being watched are `kube-flannel-cfg`, `kube-proxy`, `kube-root-ca.crt`, and `coredns`. They of course depend on the components deployed in the cluster. Let's keep this generic.

- Verb: `watch`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `configmaps/kube-system/configmaps/{serviceaccount}`
- Namespace: `kube-system`

## 1.8 Nodes creating tokens

### 1.8.1 Nodes creating tokens for SAs in the kube-system namespace

Since nodes watch the configmaps of the Pods they are running, as the tokens of their service accounts expire, they will recreate them.

- Verb: `create`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `/api/v1/namespaces/kube-system/serviceaccounts/{serviceaccount}/token`

## 1.9 Nodes getting their own status

Nodes poll their own status every ten seconds.

- Verb: `get`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `/api/v1/nodes/{the same node}`

## 1.10 Nodes patching their status to update conditions

Nodes update their status (e.g., `MemoryPressure`, `DiskPressure`, `PIDPressure`) by patching their own status every five minutes.

- Verb: `patch`
- Users: `system:node:.*`, groups `["system:nodes","system:authenticated"]`
- Objects: `/api/v1/nodes/{the same node}`

## 1.11 CoreDNS watching namespaces

- Verb: `watch`
- Users: `system:serviceaccount:kube-system:coredns`
- Objects: `namespaces`

## 1.12 Kube-proxy watching nodes

- Verb: `watch`
- Users: `system:serviceaccount:kube-system:kube-proxy`
- Objects: `nodes`

# 2 Log setups level 1

## 2.1 Namespace log N1 - Simple namespace creation and deletion

- Create a namespace:

```
kubectl create namespace rising
```

- List the namespaces:

```
kubectl get namespaces
```

- Get the namespace using `-o yaml`:

```
kubectl get namespace rising -o yaml
```

- Describe the namespace:

```
kubectl describe namespace rising
```

- Delete the namespace:

```
kubectl delete namespace rising
```

## 2.2 Namespace log N2 - Namespace with resource quotas

- **Prerequisite:** recreate the rising namespace:

```
kubectl create namespace rising
```

- Create a ResourceQuota for the namespace:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ResourceQuota
  metadata:
    name: rising-quota
  spec:
    hard:
      pods: "1"
      requests.cpu: "1"
      requests.memory: 1Gi
      limits.cpu: "2"
      limits.memory: 2Gi
EOF
```

- Inspect the ResourceQuota:

```
kubectl -n rising describe resourcequota rising-quota
```

- Create a Pod that will exceed the quota:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: over-quota-pod
    namespace: rising
  spec:
    containers:
      - name: over-quota-container
        image: nginx
        resources:
          requests:
            cpu: "1"
            memory: 1.5Gi
          limits:
            cpu: "2"
            memory: 2Gi
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Now create a Pod that will respect the quota:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: within-quota-pod
    namespace: rising
  spec:
    containers:
      - name: within-quota-container
        image: nginx
        resources:
          requests:
            cpu: "1"
            memory: 1Gi
          limits:
            cpu: "2"
            memory: 2Gi
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Run a further Pod, it will fail:

```
kubectl -n rising run --image=nginx nginx
```

- Clean everything up, one pod will fail because it was never created:

```
kubectl -n rising delete pod over-quota-pod
kubectl -n rising delete pod within-quota-pod
kubectl delete resourcequota rising-quota
```

## 2.3 Pod log P1 - Simple Pod creation and inspection

- Create a pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: rising-pod
    namespace: rising
  spec:
    containers:
      - name: first-container
        image: nginx
      - name: second-container
        image: alpine
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod rising-pod
```

- Patch the Pod and try to add a third container, this will fail:

```
kubectl -n rising patch pod rising-pod --type='json' -p='[{"op": "add", "path":
↪ "/spec/containers/1", "value": {"name": "third-container", "image": "nginx"}}]'
```

- Get the pod log:

```
kubectl -n rising logs rising-pod -c second-container
```

- Execute commands in the second containers, the first will complain about the shell, the second will work:

```
kubectl -n rising exec -t rising-pod -c second-container -- /bin/bash -c "echo Hello again,
↳ Kubernetes!"
kubectl -n rising exec -t rising-pod -c second-container -- /bin/sh -c "echo Hello again,
↳ Kubernetes!"
```

- Delete the pod:

```
kubectl -n rising delete pod rising-pod
```

## 2.4 Pod log P2 - Image download

- **Prerequisite:** run the following helper script to wipe the image cache on all nodes:

```
for node in $(kubectl get nodes -o jsonpath='{.items[*].metadata.name}'); do
  echo "Pruning images on $node"
  ssh -F ssh/cluster-setup $node "sudo crictl rmi --prune"
done
```

- Create a Pod with a container that uses a non-existent image:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: image-pod
    namespace: rising
  spec:
    containers:
      - name: image-container
        image: www.fbk.eu/non-existent-image
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Get the pod log:

```
kubectl -n rising logs image-pod -c image-container
```

- Get the events that led to the failure:

```
kubectl get events -n rising --sort-by='.metadata.creationTimestamp' --selector
↳ 'involvedObject.name=image-pod'
```

- Delete the pod:

```
kubectl -n rising delete pod image-pod
```

- Create a pod with an image that will surely not be in cache:



```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: image-pod
    namespace: rising
  spec:
    containers:
      - name: image-container
        image: busybox
        command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Wait for the Pod to be downloaded and monitor the events:

```
kubectl wait --for=condition=Ready pod/image-pod -n rising
kubectl get events -n rising --sort-by='.metadata.creationTimestamp'
```

- Delete the pod:

```
kubectl -n rising delete pod image-pod
```

## 2.5 ConfigMap C1 - Create a ConfigMap and use it in a Pod, then delete it

- Create a ConfigMap:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ConfigMap
  metadata:
    name: my-config
    namespace: rising
  data:
    status: "decent"
EOF
```

- Get the configmap

```
kubectl -n rising get configmap my-config -o yaml
```

- Deploy a Pod that uses the ConfigMap:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: config-pod
    namespace: rising
  spec:
    containers:
      - name: config-container
        image: alpine
        command: ["sh", "-c", 'echo Status is ${STATUS} && sleep 3600']
        env:
          - name: STATUS
            valueFrom:
              configMapKeyRef:
                name: my-config
                key: status
EOF
```

- Access the logs of the Pod:

```
kubectl wait --for=condition=Ready pod/config-pod -n rising
kubectl -n rising logs config-pod -c config-container
```

- Clean everything up:

```
kubectl -n rising delete pod config-pod
kubectl -n rising delete configmap my-config
```

## 2.6 Secret S1 - Create a Secret and use it as environment in a Pod, then delete it

- Create a secret

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Secret
  metadata:
    name: my-secret
    namespace: rising
  type: Opaque
  data:
    username: $(echo -n "admin" | base64)
    password: $(echo -n "password" | base64)
EOF
```

- Inspect the secret:

```
kubectl -n rising get secret my-secret -o yaml
```

- Attach the secret to a Pod:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: secret-pod
    namespace: rising
  spec:
    containers:
      - name: secret-container
        image: alpine
        command: ["sh", "-c", 'echo Username is \${USERNAME} && echo Password is \${PASSWORD} && sleep 3600']
        env:
          - name: USERNAME
            valueFrom:
              secretKeyRef:
                name: my-secret
                key: username
          - name: PASSWORD
            valueFrom:
              secretKeyRef:
                name: my-secret
                key: password
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod secret-pod
```

- Access the logs of the Pod:

```
kubectl wait --for=condition=Ready pod/secret-pod -n rising
kubectl -n rising logs secret-pod -c secret-container
```

- Clean everything up:

```
kubectl -n rising delete pod secret-pod
kubectl -n rising delete secret my-secret
```

## 2.7 Secret S2 - Create a Secret and mount it in a Pod

- Create a secret

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Secret
  metadata:
    name: my-secret-2
    namespace: rising
  type: Opaque
  data:
    username: $(echo -n "admin" | base64)
    password: $(echo -n "password" | base64)
EOF
```

- Create a Pod, but now, mount the secret as a volume:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: secret-pod-2
    namespace: rising
  spec:
    containers:
      - name: secret-container-2
        image: alpine
        command: ["sh", "-c", 'echo Username is \${USERNAME} && echo Password is \${PASSWORD} && sleep 3600']
        volumeMounts:
          - name: secret-volume
            mountPath: /etc/secret
    volumes:
      - name: secret-volume
        secret:
          secretName: my-secret-2
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod secret-pod-2
```

- Access the Secret from the Pod:

```
kubectl wait --for=condition=Ready pod/secret-pod-2 -n rising
kubectl -n rising exec secret-pod-2 -- cat /etc/secret/username
echo
kubectl -n rising exec secret-pod-2 -- cat /etc/secret/password
```

- Clean everything up:

```
kubectl -n rising delete pod secret-pod-2
kubectl -n rising delete secret my-secret-2
```

## 2.8 Node log N01 - Node inspection

- Assess the available nodes in the cluster:

```
kubectl get nodes
```

- Inspect one of the nodes:

```
kubectl describe node kubeadm-worker1
```

## 2.9 Certificate log CSR1 - Create a CertificateSigningRequest and approve it

- Note: an helper script is available in `scripts/generate_k8s_user.sh` that automates what is done in this section.
- Use openssl to generate a key and a certificate signing request:

```
export TEMPDIR=$(mktemp -d)
export ORIGINDIR=$(pwd)
cd $TEMPDIR
export KUBEUSER=malicious-user
openssl genpkey -out ${KUBEUSER}.key -algorithm Ed25519
openssl req -new -key ${KUBEUSER}.key -out ${KUBEUSER}.csr -subj
↳ "/CN=${KUBEUSER}/O=rising/O=fbk/"
echo "Key created, available in ${TEMPDIR} for user ${KUBEUSER}"
```

- Create a CertificateSigningRequest:

```
cat <<EOF | kubectl apply -f -
  apiVersion: certificates.k8s.io/v1
  kind: CertificateSigningRequest
  metadata:
    name: ${KUBEUSER}
  spec:
    request: $(cat ${KUBEUSER}.csr | base64 | tr -d '\n')
    signerName: kubernetes.io/kube-apiserver-client
    usages:
      - client auth
EOF
```

- Deny the CertificateSigningRequest:

```
kubectl certificate deny ${KUBEUSER}
```

- Delete the failed CSR and generate a new keypair for a legitimate user:

```
rm ${KUBEUSER}.csr ${KUBEUSER}.key
kubectl delete csr ${KUBEUSER}
export KUBEUSER=rising-user
openssl genpkey -out ${KUBEUSER}.key -algorithm Ed25519
openssl req -new -key ${KUBEUSER}.key -out ${KUBEUSER}.csr -subj
↳ "/CN=${KUBEUSER}/O=rising/O=fbk/"
echo "Key created, available in ${TEMPDIR} for user ${KUBEUSER}"
```

- Generate a new CertificateSigningRequest:

```
cat <<EOF | kubectl apply -f -
  apiVersion: certificates.k8s.io/v1
  kind: CertificateSigningRequest
  metadata:
    name: ${KUBEUSER}
  spec:
    request: $(cat ${KUBEUSER}.csr | base64 | tr -d '\n')
    signerName: kubernetes.io/kube-apiserver-client
    usages:
      - client auth
EOF
```

- Inspect the status of the CSR:

```
kubectl get csr ${KUBEUSER} -o yaml
```

- Approve the CertificateSigningRequest:

```
kubectl certificate approve ${KUBEUSER}
```

- Obtain the certificate from the CSR and store it in the folder

```
kubectl get csr ${KUBEUSER} -o jsonpath='{.status.certificate}' | base64 -d > ${KUBEUSER}.cert
head -n 2 ${KUBEUSER}.cert
```

- Use the certificate to generate a .kubeconfig for the user:

```
cat <<EOF > ${KUBEUSER}.kubeconfig
  apiVersion: v1
  kind: Config
  preferences: {}
  clusters:
    $(kubectl config view --raw --minify --flatten | yq .clusters)
  contexts: []
  users: []
EOF
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config set-credentials ${KUBEUSER}
↪ --client-certificate=${KUBEUSER}.cert --client-key=${KUBEUSER}.key --embed-certs=true
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config set-context ${KUBEUSER}
↪ --cluster=kubernetes --user=${KUBEUSER} --cluster=risinglab
kubectl --kubeconfig ${KUBEUSER}.kubeconfig config use-context ${KUBEUSER}
cp ${KUBEUSER}.kubeconfig ${ORIGINDIR}
```

## 2.10 Role log R1 - Create a Role and bind it to a User

- **Prerequisite:** complete CSR1 and obtain the credentials for the user. Make sure they are available as rising-user.kubeconfig in the current directory.
- Create a Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: rising-reader
    namespace: rising
  rules:
    - apiGroups: [""]
      resources: ["pods"]
      verbs: ["get", "list"]
EOF
```

- List the roles in the namespace:

```
kubectl -n rising get roles
```

- We changed idea: let's also give `get`, `list`, `patch` and `watch` for Secrets:

```
kubectl patch role rising-reader -n rising --type='json' -p='[{"op": "add", "path":  
  ↪ "/rules/0", "value": {"apiGroups": [""], "resources": ["secrets"], "verbs": ["get",  
  ↪ "list", "patch", "watch"]}]']'
```

- Let's create a RoleBinding and assign it to the user:

```
cat <<EOF | kubectl apply -f -  
  apiVersion: rbac.authorization.k8s.io/v1  
  kind: RoleBinding  
  metadata:  
    name: pod-reader-binding  
    namespace: rising  
  subjects:  
  - kind: User  
    name: rising-user  
  roleRef:  
    kind: Role  
    name: rising-reader  
    apiGroup: rbac.authorization.k8s.io  
EOF
```

- Test the permissions of the user:

```
kubectl --kubeconfig rising-user.kubeconfig get pods -n rising  
kubectl --kubeconfig rising-user.kubeconfig get secrets -n rising  
kubectl --kubeconfig rising-user.kubeconfig delete pod non-existent-pod -n rising
```

## 2.11 Role log R2 - Create a ClusterRole and bind it to a Group

- **Prerequisite:** complete CSR1 and obtain the credentials for the user. Make sure they are available as `rising-user.kubeconfig` in the current directory.
- **Prerequisite:** complete R1 and keep the `rising-reader` Role.
- Create a ClusterRole:

```
cat <<EOF | kubectl apply -f -  
  apiVersion: rbac.authorization.k8s.io/v1  
  kind: ClusterRole  
  metadata:  
    name: cluster-reader  
  rules:  
  - apiGroups: [""]  
    resources: ["pods"]  
    verbs: ["get", "list"]  
  - apiGroups: [""]  
    resources: ["secrets"]  
    verbs: ["get", "list", "patch", "watch"]  
EOF
```

- Actually, let's straight out replace the ClusterRole with a new one:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: cluster-reader
  rules:
  - apiGroups: [""]
    resources: ["nodes"]
    verbs: ["get", "list"]
EOF
```

- Create a ClusterRoleBinding and assign it to the fbk group:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRoleBinding
  metadata:
    name: cluster-reader-binding
  subjects:
  - kind: Group
    name: fbk
  roleRef:
    kind: ClusterRole
    name: cluster-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Test the permissions of the user:

```
kubectl --kubeconfig rising-user.kubeconfig get nodes
kubectl --kubeconfig rising-user.kubeconfig get pods -n rising
kubectl --kubeconfig rising-user.kubeconfig get configmaps -n rising
kubectl --kubeconfig rising-user.kubeconfig get pods -n kube-system
```

- Clean everything up:

```
kubectl delete rolebinding pod-reader-binding -n rising
kubectl delete role rising-reader -n rising
kubectl delete clusterrolebinding cluster-reader-binding
kubectl delete clusterrole cluster-reader
kubectl delete csr rising-user
```

## 2.12 Role log R3 - Create a ServiceAccount for a long-running Pod and bind it to a Role

- Set up a ServiceAccount:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: long-running-sa
    namespace: rising
EOF
```

- Create a Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: long-running-role
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
EOF
```

- Bind the ServiceAccount to the Role:

```
cat <<EOF | kubectl apply -f -
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: long-running-role-binding
    namespace: rising
  subjects:
  - kind: ServiceAccount
    name: long-running-sa
    namespace: rising
  roleRef:
    kind: Role
    name: long-running-role
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Create a Pod that uses the ServiceAccount:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Pod
  metadata:
    name: long-running-pod
    namespace: rising
  spec:
    serviceAccountName: long-running-sa
    containers:
    - name: long-running-container
      image: ubuntu
      command: ["sh", "-c", "echo Hello, Kubernetes! && sleep 3600"]
EOF
```

- Inspect the Pod:

```
kubectl -n rising describe pod long-running-pod | head -n 10
```

- Try to use the ServiceAccount from inside the Pod:

```
kubectl wait --for=condition=Ready pod/long-running-pod -n rising
kubectl -n rising exec long-running-pod -- sh -c 'apt-get -qq update; apt-get install -yqq
  curl yq >/dev/null;
  export KUBE_TOKEN=$(cat /var/run/secrets/kubernetes.io/serviceaccount/token);
  curl -sSk -H "Authorization: Bearer $KUBE_TOKEN"
  "https://$KUBERNETES_SERVICE_HOST:$KUBERNETES_PORT_443_TCP_PORT/apis/authentication.k8s.io/v1/selfsubject
  > temp.json; jq .message temp.json
  curl -sSk -H "Authorization: Bearer $KUBE_TOKEN"
  "https://$KUBERNETES_SERVICE_HOST:$KUBERNETES_PORT_443_TCP_PORT/api/v1/namespaces/rising/pods"
  > temp.json; head -n 5 temp.json'
```



- Clean everything up:

```
kubectl -n rising delete pod long-running-pod
kubectl -n rising delete role long-running-role
kubectl -n rising delete rolebinding long-running-role-binding
kubectl -n rising delete serviceaccount long-running-sa
```

## 2.13 Access Review AR1 - Use can-i to view permissions of self, other users, over different namespaces

- Create a set of ServiceAccounts for the logs (while this should not be the case in a real-world scenario, it is done since SAs and Users are equivalent once authorized):

```
kubectl apply -f - <<EOF
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: alice
    namespace: rising
---
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: bob
    namespace: rising
---
  apiVersion: v1
  kind: ServiceAccount
  metadata:
    name: charlie
    namespace: rising
EOF
```

- Create tokens for the ServiceAccounts:

```
kubectl apply -f - <<EOF
  apiVersion: v1
  kind: Secret
  metadata:
    name: alice-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: alice
  type: kubernetes.io/service-account-token
---
  apiVersion: v1
  kind: Secret
  metadata:
    name: bob-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: bob
  type: kubernetes.io/service-account-token
---
  apiVersion: v1
  kind: Secret
  metadata:
    name: charlie-token
    namespace: rising
    annotations:
      kubernetes.io/service-account.name: charlie
  type: kubernetes.io/service-account-token
---
EOF
```

- Create two Roles, one for reading Pods and one for reading Secrets:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: pod-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["pods"]
    verbs: ["get", "list"]
---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    name: secret-reader
    namespace: rising
  rules:
  - apiGroups: [""]
    resources: ["secrets"]
    verbs: ["get", "list"]
EOF
```

- Bind the ServiceAccounts to the Roles:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: pod-reader-binding
    namespace: rising
  subjects:
    - kind: ServiceAccount
      name: alice
      namespace: rising
    - kind: ServiceAccount
      name: bob
      namespace: rising
  roleRef:
    kind: Role
    name: pod-reader
    apiGroup: rbac.authorization.k8s.io
---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: RoleBinding
  metadata:
    name: secret-reader-binding
    namespace: rising
  subjects:
    - kind: ServiceAccount
      name: charlie
      namespace: rising
  roleRef:
    kind: Role
    name: secret-reader
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Obtain credentials for the ServiceAccounts and generate a kubeconfig for them:

```
kubectl get secret alice-token -n rising -o jsonpath='{.data.token}' | base64 -d > alice.token
kubectl get secret bob-token -n rising -o jsonpath='{.data.token}' | base64 -d > bob.token
kubectl get secret charlie-token -n rising -o jsonpath='{.data.token}' | base64 -d >
↳ charlie.token
cat <<EOF > alice.kubeconfig
  apiVersion: v1
  kind: Config
  preferences: {}
  clusters:
    $(kubectl config view --raw --minify --flatten | jq .clusters)
  contexts: []
  users: []
  EOF
cp alice.kubeconfig bob.kubeconfig
cp alice.kubeconfig charlie.kubeconfig
kubectl --kubeconfig alice.kubeconfig config set-credentials alice --token=$(cat alice.token)
kubectl --kubeconfig bob.kubeconfig config set-credentials bob --token=$(cat bob.token)
kubectl --kubeconfig charlie.kubeconfig config set-credentials charlie --token=$(cat
↳ charlie.token)
kubectl --kubeconfig alice.kubeconfig config set-context alice --cluster=kubernetes
↳ --user=alice --cluster=risinglab
kubectl --kubeconfig bob.kubeconfig config set-context bob --cluster=kubernetes --user=bob
↳ --cluster=risinglab
kubectl --kubeconfig charlie.kubeconfig config set-context charlie --cluster=kubernetes
↳ --user=charlie --cluster=risinglab
kubectl --kubeconfig alice.kubeconfig config use-context alice
kubectl --kubeconfig bob.kubeconfig config use-context bob
kubectl --kubeconfig charlie.kubeconfig config use-context charlie
```

```
rm alice.token bob.token charlie.token
```

- Check as Alice if she can list Pods and Secrets in the namespace:

```
kubectl --kubeconfig alice.kubeconfig auth can-i list pods -n rising
kubectl --kubeconfig alice.kubeconfig auth can-i list secrets -n rising
```

- Check as Bob if he can list Pods in the namespaces **rising** and **kube-system**:

```
kubectl --kubeconfig bob.kubeconfig auth can-i list pods -n rising
kubectl --kubeconfig bob.kubeconfig auth can-i list pods -n kube-system
```

- Check as Charlie if he can list Secrets in the namespace:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i list secrets -n rising
```

- Check as Charlie if he can list nodes:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i list nodes
```

- Check what Charlie can do in the **rising** namespace:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i --list -n rising
```

## 2.14 Access Review AR2 - Use auth whoami to view the current user

- **Prerequisite:** complete AR1 and obtain the credentials for the users. Make sure they are available as **alice.kubeconfig**, **bob.kubeconfig** and **charlie.kubeconfig** in the current directory.
- Check as each of the users who they are:

```
kubectl --kubeconfig alice.kubeconfig auth whoami
kubectl --kubeconfig bob.kubeconfig auth whoami
kubectl --kubeconfig charlie.kubeconfig auth whoami
```

## 2.15 Access Review AR3 - Impersonating a user

- **Prerequisite:** complete AR1 and obtain the credentials for the users. Make sure they are available as **alice.kubeconfig**, **bob.kubeconfig** and **charlie.kubeconfig** in the current directory.
- Augment Charlie's capabilities to let him impersonate ServiceAccounts:

```
kubectl apply -f - <<EOF
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRole
  metadata:
    name: impersonator
  rules:
    - apiGroups: [""]
      resources: ["serviceaccounts"]
      verbs: ["impersonate"]
      resourceNames: ["alice", "bob"]
  ---
  apiVersion: rbac.authorization.k8s.io/v1
  kind: ClusterRoleBinding
  metadata:
    name: impersonator-binding
  subjects:
    - kind: ServiceAccount
      name: charlie
      namespace: rising
  roleRef:
    kind: ClusterRole
    name: impersonator
    apiGroup: rbac.authorization.k8s.io
EOF
```

- Check if Charlie can impersonate any user or SA:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i impersonate users
kubectl --kubeconfig charlie.kubeconfig auth can-i impersonate serviceaccounts
```

- Let Charlie see if Alice can list Pods in **rising**, and what she can do in **kube-system**:

```
kubectl --kubeconfig charlie.kubeconfig auth can-i list pods -n rising --as
↳ system:serviceaccount:rising:alice
kubectl --kubeconfig charlie.kubeconfig auth can-i --list -n kube-system --as
↳ system:serviceaccount:rising:alice
```

- As Charlie, let's impersonate Bob and create a Pod in the **rising** namespace, then list Pods:

```
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:bob run
↳ --image=ubuntu --restart=Never --namespace=rising --command -- sleep 3600
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:bob get pods -n
↳ rising
```

- As Charlie, let's impersonate Alice and try to list nodes:

```
kubectl --kubeconfig charlie.kubeconfig --as system:serviceaccount:rising:alice get nodes
```

- Clean everything up:

```
kubectl delete clusterrolebinding impersonator-binding
kubectl delete clusterrole impersonator
kubectl delete rolebinding pod-reader-binding -n rising
kubectl delete role pod-reader -n rising
kubectl delete rolebinding secret-reader-binding -n rising
kubectl delete role secret-reader -n rising
kubectl delete secret alice-token bob-token charlie-token -n rising
kubectl delete serviceaccount alice bob charlie -n rising
rm alice.kubeconfig bob.kubeconfig charlie.kubeconfig
```

## 3 Log setups level 2

### 3.1 Deployment log D1 - Create a Deployment for a ReplicaSet and expose it as a Service

- Create a Deployment:

```
cat <<EOF | kubectl apply -f -
apiVersion: apps/v1
kind: Deployment
metadata:
  name: test-deployment
  namespace: rising
  labels:
    app: test
spec:
  replicas: 2
  selector:
    matchLabels:
      app: test
  template:
    metadata:
      labels:
        app: test
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
EOF
```

- Inspect the Deployment:

```
kubectl get deployments test-deployment -n rising
```

- Inspect the rollout status:

```
kubectl rollout status deployment test-deployment -n rising
```

- Inspect the ReplicaSet created by the Deployment:

```
kubectl get replicaset -l app=test -n rising
```

- Describe the ReplicaSet directly:

```
kubectl describe replicaset -l app=test -n rising
```

- Expose the Deployment as a Service:

```
kubectl expose deployment test-deployment --type=NodePort --port=80 --target-port=80
↪ --name=test-deployment -n rising
```

- Inspect the Service:

```
kubectl get services test-deployment -n rising
```

- Tear everything down:

```
kubectl delete service test-deployment -n rising
kubectl delete deployment test-deployment -n rising
```

### 3.2 Deployment log D2 - Scale a Deployment and update its image

- Create a Deployment again:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: test-deployment
    namespace: rising
    labels:
      app: test
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: test
    template:
      metadata:
        labels:
          app: test
      spec:
        containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
EOF
```

- Scale the Deployment to 3 replicas:

```
kubectl scale deployment test-deployment --replicas=3 -n rising
```

- Inspect the Deployment:

```
kubectl get deployments test-deployment -n rising
```

- Update the image of the Deployment:

```
kubectl set image deployment test-deployment nginx=nginx:1.16.1 -n rising
```

- Inspect the rollout status:

```
kubectl rollout status deployment test-deployment -n rising
```

- Scale down the Deployment to 2 replicas:

```
kubectl scale deployment test-deployment --replicas=2 -n rising
```

- Inspect ReplicaSets:

```
kubectl get replicaset -l app=test -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=test -n rising
```

- Tear everything down:

```
kubectl delete deployment test-deployment -n rising
```

### 3.3 Deployment log D3 - Roll back, pause and resume a Deployment

- Create a Deployment again:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: test-deployment
    namespace: rising
    labels:
      app: test
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: test
    template:
      metadata:
        labels:
          app: test
      spec:
        containers:
          - name: nginx
            image: nginx:latest
            ports:
              - containerPort: 80
EOF
```

- Wait for the Deployment to be ready:

```
kubectl wait --for=condition=Available deployment/test-deployment -n rising
```

- Pause the Deployment:

```
kubectl rollout pause deployment test-deployment -n rising
```

- Update the image of the Deployment:

```
kubectl set image deployment test-deployment nginx=nginx:1.16.1 -n rising
```

- Inspect the rollout history:

```
kubectl rollout history deployment test-deployment -n rising
```

- Inspect the ReplicaSets:

```
kubectl get replicaset -l app=test -n rising
```

- Resume the Deployment:

```
kubectl rollout resume deployment test-deployment -n rising
```

- Inspect the rollout history again:

```
kubectl rollout history deployment test-deployment -n rising
```

- Roll back the Deployment to the previous revision:

```
kubectl rollout undo deployment test-deployment -n rising
```

- Inspect the rollout history one more time:



```
kubectl rollout history deployment test-deployment -n rising
```

- List Pods:

```
kubectl get pods -l app=test
```

- Tear everything down:

```
kubectl delete deployment test-deployment -n rising
```

### 3.4 Deployment log D4 - Watch a Deployment fail and recover

- Install a restrictive ResourceQuota:

```
cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: ResourceQuota
  metadata:
    name: test-quota
    namespace: rising
  spec:
    hard:
      pods: "1"
EOF
```

- Create a Deployment again:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: test-deployment
    namespace: rising
    labels:
      app: test
  spec:
    replicas: 2
    selector:
      matchLabels:
        app: test
    template:
      metadata:
        labels:
          app: test
      spec:
        containers:
          - name: nginx
            image: nginx:latest
            ports:
              - containerPort: 80
EOF
```

- Check the status of the Deployment:

```
kubectl describe deployment test-deployment -n rising
```

- Relax the ResourceQuota:

```
kubectl patch resourcequota test-quota -p '{"spec":{"hard":{"pods":"5"}}}' -n rising
```

- Check the status of the Deployment again:

```
kubectl rollout status deployment test-deployment -n rising
```

- Change the image of the Deployment and put a non-existent image:

```
kubectl set image deployment test-deployment nginx=engincs:ahahahah -n rising
```

- Check the status of the Deployment:

```
kubectl rollout status deployment test-deployment -n rising
```

- Fix the image of the Deployment:

```
kubectl set image deployment test-deployment nginx=nginx:1.16.1 -n rising
```

- Now, set an impossible affinity for the Deployment:

```
kubectl patch deployment test-deployment -p
↪ '{"spec":{"template":{"spec":{"affinity":{"nodeAffinity":{"requiredDuringSchedulingIgnoredDuringExecution":{"nodeSelectorTerms":[{"matchExpressions":[{"key":"node.kubernetes.io/instance-type","operator":"NotIn","values":["worker"]}]}}}}}}}'
↪ -n rising
```

- Check the status of the Deployment:

```
kubectl rollout status deployment test-deployment -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=test -n rising
```

- Tear everything down:

```
kubectl delete deployment test-deployment -n rising
kubectl delete resourcequota test-quota -n rising
```

### 3.5 ReplicaSet log RS1 - Use bare ReplicaSets to manage Pods

- Create a ReplicaSet:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: ReplicaSet
  metadata:
    name: frontend
    namespace: rising
    labels:
      app: guestbook
      tier: frontend
  spec:
    replicas: 3
    selector:
      matchLabels:
        tier: frontend
    template:
      metadata:
        labels:
          tier: frontend
      spec:
        containers:
          - name: outward
            image: httpd:latest
EOF
```

- List the ReplicaSets:

```
kubectl get replicaset -n rising
```

- Inspect the ReplicaSet:

```
kubectl describe replicaset frontend -n rising
```

- Scale down the ReplicaSet:

```
kubectl scale replicaset frontend --replicas=2 -n rising
```

- Inspect the Pods:

```
kubectl get pods -l tier=frontend -n rising
```

- Tear everything down:

```
kubectl delete replicaset frontend -n rising
```

### 3.6 DaemonSet log DS1 - Manage a DaemonSet

- Create a DaemonSet:

```
cat <<EOF | kubectl apply -f -
  apiVersion: apps/v1
  kind: DaemonSet
  metadata:
    name: fluentd-elasticsearch
    namespace: kube-system
    labels:
      k8s-app: fluentd-logging
  spec:
    selector:
      matchLabels:
        name: fluentd-elasticsearch
    template:
      metadata:
        labels:
          name: fluentd-elasticsearch
      spec:
        tolerations:
          - key: node-role.kubernetes.io/control-plane
            operator: Exists
            effect: NoSchedule
        containers:
          - name: fluentd-elasticsearch
            image: quay.io/fluentd_elasticsearch/fluentd:v2.5.2
            resources:
              limits:
                memory: 200Mi
              requests:
                cpu: 100m
                memory: 200Mi
            volumeMounts:
              - name: varlog
                mountPath: /var/log
        terminationGracePeriodSeconds: 30
        volumes:
          - name: varlog
            hostPath:
              path: /var/log
EOF
```

- Inspect the DaemonSet:

```
kubectl get daemonsets fluentd-elasticsearch -n kube-system
```

- Inspect the Pods:

```
kubectl get pods -l name=fluentd-elasticsearch -n kube-system
```

- Cordon kubeadm-worker1:

```
kubectl cordon kubeadm-worker1
```

- Check the Pods again:

```
kubectl get pods -l name=fluentd-elasticsearch -n kube-system
```

- Uncordon and add a label to kubeadm-worker1:

```
kubectl uncordon kubeadm-worker1
kubectl taint nodes kubeadm-worker1 test-label=no-schedule:NoExecute
```

- Edit the DaemonSet to prevent it from running on nodes with the label:

```
kubectl patch daemonset fluentd-elasticsearch -p
↪ '{"spec":{"template":{"spec":{"tolerations":[{"key":"test-label","operator":"NotExists","effect":"NoExecu
↪ -n kube-system
```

- Check the Pods again:

```
kubectl get pods -l name=fluentd-elasticsearch -n kube-system
```

- Clean everything up:

```
kubectl delete daemonset fluentd-elasticsearch -n kube-system
```

### 3.7 Job log J1 - Manage a Job

- Create a Job:

```
cat <<EOF | kubectl apply -f -
apiVersion: batch/v1
kind: Job
metadata:
  name: pi
  namespace: rising
spec:
  template:
    spec:
      containers:
      - name: pi
        image: perl:5.34.0
        command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(2000)"]
        restartPolicy: Never
      backoffLimit: 4
EOF
```

- List available Jobs:

```
kubectl get jobs -n rising
```

- Inspect the Job:

```
kubectl describe jobs pi -n rising
```

- Check the Pods:

```
kubectl get pods -l job-name=pi -n rising
```

- Assess the logs generated by the Job:

```
kubectl logs jobs pi -n rising
```

- Update the spec of the Job to trigger auto deletion:

```
kubectl patch job pi -p '{"spec":{"ttlSecondsAfterFinished":60}}' -n rising
```

- Wait for the Job to be deleted:

```
kubectl wait --for=delete job/pi -n rising
```

### 3.8 CronJob log CJ1 - Manage a CronJob

- Create a CronJob:

```
cat <<EOF | kubectl apply -f -
  apiVersion: batch/v1
  kind: CronJob
  metadata:
    name: hello
    labels:
      cj: hello
    namespace: rising
  spec:
    schedule: "* * * * *"
    jobTemplate:
      spec:
        template:
          spec:
            containers:
            - name: hello
              image: busybox:1.28
              imagePullPolicy: IfNotPresent
              command:
              - /bin/sh
              - -c
                date; echo Hello from the Kubernetes cluster
            restartPolicy: OnFailure
EOF
```

- List available CronJobs:

```
kubectl get cronjobs -n rising
```

- Inspect the CronJob:

```
kubectl describe cronjobs hello -n rising
```

- Wait two-three minutes and check the Jobs created by the CronJob:

```
sleep 120
kubectl get jobs -l cj=hello -n rising
```

- Clean everything up:

```
kubectl delete cronjob hello -n rising
kubectl delete jobs -l cj=hello -n rising
```

## 4 Log setups level 3

### 4.1 Stateful Set log SS1 - Manage a StatefulSet

- Create a dumb StorageClass for the StatefulSet:

```
cat <<EOF | kubectl apply -f -
  apiVersion: storage.k8s.io/v1
  kind: StorageClass
  metadata:
    name: local-storage
    annotations:
      storageclass.kubernetes.io/is-default-class: "true"
  provisioner: kubernetes.io/no-provisioner
  volumeBindingMode: WaitForFirstConsumer
EOF
```

- Create some PersistentVolumes:

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: PersistentVolume
metadata:
  name: pv-0
spec:
  capacity:
    storage: 1Gi
  volumeMode: Filesystem
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  storageClassName: local-storage
  local:
EOF
```

The PersistentVolume "pv-0" is invalid: spec: Required value: must specify a volume type

- Create a StatefulSet with an associated Service:

```

cat <<EOF | kubectl apply -f -
  apiVersion: v1
  kind: Service
  metadata:
    name: nginx
    namespace: rising
    labels:
      app: nginx
  spec:
    ports:
      - port: 80
        name: web
    clusterIP: None
    selector:
      app: nginx
---
  apiVersion: apps/v1
  kind: StatefulSet
  metadata:
    namespace: rising
    name: web
  spec:
    selector:
      matchLabels:
        app: nginx
    serviceName: "nginx"
    replicas: 3
    minReadySeconds: 10 # by default is 0
    template:
      metadata:
        labels:
          app: nginx
      spec:
        terminationGracePeriodSeconds: 10
        containers:
          - name: nginx
            image: registry.k8s.io/nginx-slim:0.24
            ports:
              - containerPort: 80
                name: web
            volumeMounts:
              - name: www
                mountPath: /usr/share/nginx/html
        volumeClaimTemplates:
          - metadata:
              name: www
            spec:
              accessModes: [ "ReadWriteOnce" ]
              resources:
                requests:
                  storage: 200Mi
EOF

```

- Inspect the StatefulSet:

```
kubectl get statefulsets web -n rising
```

- Inspect the Pods:

```
kubectl get pods -l app=nginx -n rising
```

- Scale up the StatefulSet:



```
kubectl scale statefulsets web --replicas=4 -n rising
```

- Inspect the Pods again:

```
kubectl get pods -l app=nginx -n rising
```

- Scale down to 2:

```
kubectl scale statefulsets web --replicas=2 -n rising
```